PRODUCT PROGRESS REPORT

FIC BACKROOMS

EVENT MANAGEMENT SYSTEM

Event Operations Engine | Movie & Theater Management | Box Office POS & Ticketing

FORGE INDIA CONNECT PVT. LTD SHAPING FUTURE

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Abstract

Project Overview

FIC BackRooms is a unified, multi-tenant digital management and ticketing platform engineered to streamline the full operational lifecycle of public events, academic conferences, movie screenings, and theater hall operations. The platform integrates two distinct yet harmonious operational engines:

- Event Domain: Caters to college symposiums, hackathons, corporate summits, workshops, exhibitions, and music concerts with flexible solo/group registration workflows.
- Theater & Cinema Domain: Caters to single-screen theaters, multiplexes, and auditoriums with multi-screen scheduling, interactive 2D seat matrix selection, walk-in box office POS, and Food & Beverage (F&B) concession management.

Instead of depending on disparate third-party tools such as manual spreadsheets, Google Forms, standalone ticket printers, and fragmented payment links, FIC BackRooms unifies attendee registrations, seat allocation, cashier operations, concession fulfillment, live QR entry verification, and revenue analytics into a single enterprise system.

Purpose of the Product

The primary purpose of FIC BackRooms is to eliminate administrative overhead, prevent revenue leakage, and offer a frictionless digital journey for organizers, cinema operators, staff, and attendees.

For event organizers, the product provides instant portal setup, custom form creation, ticket tier customization, speaker curation, and check-in dashboards. For theater and cinema operators, it solves critical industry challenges including real-time seat locking during high booking concurrency, synchronization between on-counter cashier sales and online ticket buyers, integrated concession pre-ordering, and rapid usher gate verification.

Core Multi-Domain Capabilities

FIC BackRooms bridges event registration and cinema ticketing through tailored modules:

- **Flexible Event Registration**: Support for individual participants, team leaders with member rosters, delegate profiles, waitlists, and manual or automated approval pipelines.
- **Interactive Visual Seat Selection**: Intuitive 2D screen layout rendering customizable seat tiers (Silver, Gold, Platinum, VIP, Recliner) with temporary seat locks during checkout.
- **Dual-Channel Box Office & Online Sales**: High-speed counter POS for walk-in audiences alongside public web discovery and mobile ticketing.
- **Integrated Concessions (F&B)**: Advance snack and beverage ordering during ticket checkout, backed by a real-time Kitchen Display System (KDS).
- **Sub-Second QR Gate Verification**: Mobile-ready check-in apps for event volunteers and theater ushers with immediate duplicate entry detection.

Role-Based Access Architecture

FIC BackRooms employs a granular Role-Based Access Control (RBAC) model to ensure that every stakeholder has specialized views, dedicated toolsets, and strict permission boundaries:

Role	Domain	Responsibilities & Platform Capabilities
Super Admin	Platform Level	Manages portal tenants, approves theater partner signups, configures global settings, and monitors platform-wide financial transactions.
Event Organizer	Events	Creates events, sets agenda/schedules, configures ticket classes and registration questionnaires, approves delegates, and tracks event sales.
Event Staff / Volunteer	Events	Executes day-of-event operations, scans QR tickets at venue entry gates, validates participant attendance, and assists attendees.
Theater Manager	Theater	Manages screens and auditoriums, configures seat matrices and pricing tiers, curates movie catalogs, schedules daily showtimes, and edits the F&B menu.
Theater Cashier	Theater	Operates the walk-in Box Office POS terminal, selects available seats in real-time, collects cash/card payments, and prints physical/digital tickets.
Theater Usher	Theater	Performs rapid camera QR ticket scanning at auditorium doors, validates showtime and seat rows, and flags invalid or duplicate entries.
F&B Concession Staff	Theater	Views incoming snack orders via the Kitchen Display System (KDS), manages inventory status, and marks orders as prepared and picked up.
Public User / Customer	Unified	Discovers upcoming events and movies, browses showtimes and trailers, selects seats, pre-orders snacks, and accesses digital tickets with QR passes.

Security & Workflow Clarity

This compartmentalized architecture prevents unauthorized operational overlaps. Cashiers have fast, uncluttered interfaces focused purely on rapid ticket dispensing; ushers have lightweight mobile screens optimized for scanning throughput; and theater managers retain complete control over pricing, movies, and auditorium configurations without exposing administrative platform settings.

Introduction

Modern audience engagement and venue management require sophisticated digital infrastructure. Whether managing a 1,500-participant university symposium or running daily prime-time movie screenings across multiple cinema screens, organizations face high expectations for rapid ticketing, transparent seating, and smooth admissions.

Traditionally, event planning and cinema ticketing have existed in silos. Event management frequently relies on ad-hoc spreadsheets, paper check-lists, and generic form builders. Concurrently, regional single-screen cinemas and mid-sized auditoriums are constrained by outdated legacy desktop POS software that cannot sync with modern web booking portals or handle concession pre-orders.

FIC BackRooms resolves these limitations by offering a modern, cloud-native software suite. Developed by *Forge India Connect Pvt Ltd*, the platform brings enterprise-grade event registration, cinema hall seat booking, counter box-office POS, concession management, and QR access control together in a single coordinated solution.

Problem Statement

1. Disjointed Event Tools

Organizers often use Google Forms for data collection, WhatsApp for team alerts, and Excel sheets for attendance. This causes missing registrant data, unverified payments, and lengthy lines at entry.

2. Cinema Seat Allocation Conflicts

Theaters without real-time seat locking suffer from concurrency collisions, where counter cashiers and online users book the same seat simultaneously, causing double bookings and customer conflict.

3. Disconnected Concession (F&B) Sales

Food and snack sales in cinemas are typically managed on independent billing machines. Customers cannot order snacks with their movie tickets, causing severe counter crowding during 10-minute intermissions.

4. Ticket Duplication & Gate Latency

Manual ticket inspection or paper tearing is slow and vulnerable to forged passes or forwarded screenshots. High-capacity events and movie screenings require sub-second cryptographic validation.

5. Fragmented Revenue & Payout Accounting

Venue managers struggle to calculate exact revenue splits across ticket categories, distributor shares, concession profit margins, and internet convenience fees. Manual reconciliation between bank accounts, cash drawers, and payment gateways leads to accounting discrepancies and delayed distributor settlements.

Product Objectives

The primary objective of FIC BackRooms is to deliver a reliable, secure, and intuitive platform for multi-domain venue and audience management. Key goals include:

1. Unified Multi-Tenant Management

Provide dedicated portals where colleges, companies, independent cinema owners, and multiplex chains can operate under their own branding, configure independent staff teams, and control their commercial listings.

2. Advanced Cinema & Auditorium Seat Engine

Implement a visual screen builder that allows theater managers to create custom seating configurations (rows, columns, aisles, wheelchair spots) across multiple pricing categories (Silver, Gold, VIP, Recliner). Ensure reliable real-time seat locking with automatic expiry timers during online checkout.

3. Dual-Channel Box Office & Public Ticketing

Deliver both a high-throughput Box Office Cashier POS for rapid walk-in ticket sales and a responsive public discovery portal where users can browse events and movies, watch trailers, select seats, and purchase tickets online.

4. Integrated Concessions & Kitchen Display

Allow moviegoers to pre-order food and beverages during ticket booking or purchase snacks directly at the theater. Provide F&B staff with an automated kitchen order dashboard to streamline snack prep and pickup before showtime.

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5. Role-Optimized Mobile QR Verification

Equip event staff and theater ushers with lightweight, camera-based QR scanners to achieve sub-second ticket validation, display seat and show details immediately, and prevent duplicate entries.

6. Automated Financial Reporting & Reconciliation

Generate consolidated reports covering registration statistics, gross box office collections, distributor revenue shares, occupancy percentages, and concession profit margins.

Work Done So Far

Core development has reached a highly mature milestone, with functional implementations across both Event and Theater domains within the React frontend and Spring Boot backend:

A. Completed Event Management Modules

- **Public Event Discovery**: Comprehensive listing with category filters, search, and rich event detail pages with speaker and agenda highlights.
- **Dynamic Registration Forms**: Configurable participant inputs (solo registrations, group/team rosters, custom corporate fields).
- **Ticket Class Management**: Tiered ticket creation (Early Bird, General, VIP, Student) with quota enforcement and pricing rules.
- **Event Operations Dashboard**: Submodules for Agendas, Speakers, Sponsors, Exhibitors, Incident Reporting, and Budget management.
- **Event Day QR Check-In**: Instant badge/ticket QR scanning for volunteer teams with attendance logging.

B. Completed Theater & Cinema Modules

- **Theater Partner Onboarding & SuperAdmin**: Multi-theater registration, partner verification, and portal licensing.
- **Theater Manager Dashboard**:
  - *Screen & Auditorium Setup*: Configures multiple screens per cinema with individual seat layout templates.
  - *Movie Catalog Management*: Movie metadata curation (posters, genres, language, duration, release status, trailer URLs).
  - *Showtime Scheduling*: Maps movies to specific screens, dates, and showtimes with tier-based pricing.
  - *F&B Menu Builder*: Setup of concession categories, snacks, beverages, combo deals, and stock toggles.
- **Interactive 2D Seat Layout Engine**: Visual seat selection interface displaying real-time seat availability, aisle gaps, category pricing, and active seat hold timers.
- **Cashier Box Office POS**: Dedicated rapid interface for walk-in audiences, supporting cash/card payment recording and instant ticket generation.
- **Usher Gate Scanner**: Optimized mobile scanner displaying show confirmation, screen number, seat row, and preventing reuse.
- **Theater F&B Dashboard**: Kitchen order queue tracking incoming snack orders, preparation statuses, and counter handoffs.

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System Operational Workflows

1. Event Registration & Gate Check-In Flow

1. **Browse & Select Event**: Public user browses events, reviews agendas, and picks ticket class.
2. **Custom Form Submission**: User submits solo or group details; system checks eligibility and waitlists.
3. **Payment / Approval**: Payment processed or queued for organizer manual approval.
4. **Digital QR Pass Issued**: Encrypted QR ticket sent via email and available in user profile.
5. **Venue Check-In**: Event volunteer scans QR code, confirms registration, and marks attendance.

2. Cinema Booking, Concession & Usher Entry Flow

1. **Movie & Showtime Discovery**: User selects theater, movie, language, date, and showtime.
2. **Interactive Seat Selection & Hold**: User picks seats on the 2D matrix; seats lock for 8 minutes.
3. **Concession (F&B) Add-Ons**: User pre-selects popcorn, snacks, and beverage combos.
4. **Checkout & Ticket Generation**: Single transaction generates Movie QR Pass + F&B Token.
5. **Dual On-Site Fulfillment**:
   - *Auditorium Door*: Usher scans QR ticket to verify screen, showtime, and seat row.
   - *Concession Counter*: F&B staff scans token or calls order ID to hand over snacks.

Testing & Quality Assurance Plan

To ensure enterprise reliability, FIC BackRooms is validated under real-world stress conditions:

- **Seat Concurrency & Collision Testing**: Simulating simultaneous checkout requests for identical seats to verify that transactional locks prevent double booking.
- **Box Office POS Load Testing**: Benchmarking cashier terminal transaction speeds under continuous walk-in queue conditions.
- **Usher Scanner Throughput**: Ensuring camera QR recognition takes under 500ms even in low-light theater corridor environments.
- **F&B Kitchen Order Routing**: Validating that pre-orders reflect immediately on the F&B kitchen dashboard upon successful payment.
- **Role-Based Access Enforcement**: Verifying that cashiers, ushers, organizers, and admins cannot view or alter unauthorized endpoints.
- **Payment Gateway Webhook Reliability**: Testing payment confirmation, timeout rollbacks, and automatic seat release upon checkout cancellation.

Maintenance & Support Plan

Following commercial deployment, the platform will be maintained through structured operational protocols:

Real-Time System Monitoring

Continuous uptime tracking of Spring Boot backend services, MySQL database pools, and Redis caching layers.

Automated Database Backups

Daily encrypted snapshots and point-in-time recovery for transaction logs and ticket issuance records.

Security & Compliance Updates

Regular dependency audits, JWT token rotation, and compliance with data privacy standards.

Hardware Peripheral Support

Ongoing driver maintenance and testing for thermal ticket printers, 2D barcode scanners, and counter cash drawers.

Target Market & Customer Value

Target Sector	Key Value Delivered by FIC BackRooms
Colleges & Universities	Effortless handling of technical symposiums, hackathons, and cultural fests; group team registrations, student identity tracking, and digital certificates.
Corporate Enterprises	Professional summits, product launches, delegate credentialing, branded company portals, and real-time attendance analytics.
Single-Screen Theaters	Affordable transition from manual paper ticketing to a modern cloud POS and online seat reservation system without expensive infrastructure.
Multiplexes & Cinema Chains	Multi-screen showtime management, tier-based dynamic seat pricing, integrated F&B ordering, and distributor box office settlement reports.
Auditoriums & Drama Halls	Flexible seating charts for live theater, stand-up comedy, musical concerts, and ticketed public entertainment programs.

Revenue & Business Model

FIC BackRooms implements a versatile, hybrid business model catering to diverse commercial clients:

- **Per-Screen / Per-Portal Subscription (SaaS)**:
  - *Event Portal License*: Monthly or annual fee for academic institutions and enterprise event organizers.
  - *Cinema Screen License*: Tiered monthly software subscription per active cinema screen or auditorium.
- **Online Convenience Fee**: Nominal percentage or flat convenience fee per ticket paid by end-users during online booking.
- **F&B Transaction Fee**: Small percentage commission on advance snack orders processed through the cinema platform.
- **Hardware Integration & Setup**: One-time setup fee for cinema box office counter hardware, thermal ticket printers, and usher scanners.

Conclusion

The **FIC BackRooms** platform has successfully expanded beyond traditional event management into a comprehensive, multi-tenant ecosystem encompassing public events, academic symposiums, movie screenings, and theater hall operations.

By integrating an interactive 2D seat matrix, rapid walk-in Box Office POS, usher gate QR scanning, and concession (F&B) order fulfillment alongside its established event registration engine, the platform resolves critical operational pain points for both event organizers and cinema owners.

With complete role-based dashboards (Super Admin, Organizer, Staff, Theater Manager, Cashier, Usher, and F&B Attendant) and a production-grade technology foundation (React, Spring Boot, MySQL, and JWT Security), FIC BackRooms is positioned as a scalable, high-value solution ready for commercial rollout across colleges, enterprises, single-screen cinemas, and multiplex venues.

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